le jeu Murder Mystery

En utilisant bibliothèque de pandas

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2. Étapes

2.1 Bibliothèque import

```python
import sqlite3
import pandas as pd
from pathlib import Path
```

✓ 0.0s Python

2.2 Les Informations basiques sur les tableaux

```python
data_murder = Path.cwd()/"data"/"sql-murder-mystery.db"
conn = sqlite3.connect(data_murder)
cursor = conn.cursor()
cursor = cursor.execute("select name from sqlite_master where type='table'")
# print(cursor.fetchall())
rst = pd.read_sql("select name from sqlite_master where type='table'", conn)
rst.head(10)
```

✓ 0.0s Python

	name
0	crime_scene_report
1	drivers_license
2	facebook_event_checkin
3	interview
4	get_fit_now_member
5	get_fit_now_check_in
6	solution
7	income
8	person

2.3 Transformer les tableaux en data frame

```
df_crime_scene_report = pd.read_sql("select * from crime_scene_report", conn)
(df_crime_scene_report.head(5))
# df_crime_scene_report.info()
```

✓ 0.0s Python

	date	type	description	city
0	20180115	robbery	A Man Dressed as Spider-Man Is on a Robbery Spree	NYC
1	20180115	murder	Life? Dont talk to me about life.	Albany
2	20180115	murder	Mama, I killed a man, put a gun against his he...	Reno
3	20180215	murder	REDACTED REDACTED REDACTED	SQL City
4	20180215	murder	Someone killed the guard! He took an arrow to ...	SQL City

```
df_drivers_license = pd.read_sql("select * from drivers_license", conn  )
df_drivers_license.head(5)
```

✓ 0.0s Python

	id	age	height	eye_color	hair_color	gender	plate_number	car_make	car_model
0	100280	72	57	brown	red	male	P24L4U	Acura	MDX
1	100460	63	72	brown	brown	female	XF02T6	Cadillac	SRX
2	101029	62	74	green	green	female	VKY5KR	Scion	xB
3	101198	43	54	amber	brown	female	Y5NZ08	Nissan	Rogue
4	101255	18	79	blue	grey	female	5162Z1	Lexus	GS

```
df_facebook_event_checking = pd.read_sql("select * from facebook_event_checkin", conn)
df_facebook_event_checking.head(5)
```

✓ 0.0s Python

	person_id	event_id	event_name	date
0	28508	5880	Nudists are people who wear one-button suits.\n	20170913
1	63713	3865	but that's because it's the best book on anyth...	20171009
2	63713	3999	If Murphy's Law can go wrong, it will.\n	20170502
3	63713	6436	Old programmers never die. They just branch t...	20170926
4	82998	4470	Help a swallow land at Capistrano.\n	20171022

```python
df_interview = pd.read_sql("select * from interview", conn)
df_interview.head(5)
```

✓ 0.0s Python

	person_id	transcript
0	28508	'I deny it!' said the March Hare.\n
1	63713	\n
2	86208	way, and the whole party swam to the shore.\n
3	35267	lessons in here? Why, there's hardly room for ...
4	33856	\n

```python
df_get_fit_now_member = pd.read_sql("select * from get_fit_now_member", conn)
df_get_fit_now_member.head(5)
```

✓ 0.0s Python

	id	person_id	name	membership_start_date	membership_status
0	NL318	65076	Everette Koepke	20170926	gold
1	AOE21	39426	Noe Locascio	20171005	regular
2	2PN28	63823	Jeromy Heitschmidt	20180215	silver
3	0YJ24	80651	Waneta Wellard	20171206	gold
4	3A08L	32858	Mei Bianchin	20170401	silver

```python
df_get_fit_now_check_in = pd.read_sql("select * from get_fit_now_check_in", conn)
df_get_fit_now_check_in.head(5)
```

✓ 0.0s Python

	membership_id	check_in_date	check_in_time	check_out_time
0	NL318	20180212	329	365
1	NL318	20170811	469	920
2	NL318	20180429	506	554
3	NL318	20180128	124	759
4	NL318	20171027	418	1019

```python
df_person = pd.read_sql("select * from person", conn)
df_person.head(5)
```

✓ 0.0s Python

	id	name	license_id	address_number	address_street_name	ssn
0	10000	Christoper Peteuil	993845	624	Bankhall Ave	747714076
1	10007	Kourtney Calderwood	861794	2791	Gustavus Blvd	477972044
2	10010	Muoi Cary	385336	741	Northwestern Dr	828638512
3	10016	Era Moselle	431897	1987	Wood Glade St	614621061
4	10025	Trena Hornby	550890	276	Daws Hill Way	223877684

```python
df_income = pd.read_sql("select * from income", conn)
df_income.head(5)
```

✓ 0.0s Python

	ssn	annual_income
0	100009868	52200
1	100169584	64500
2	100300433	74400
3	100355733	35900
4	100366269	73000

2.4 Trouver les informations des suspectes

```python
#step 2
df_person.head(5)
suspect_id_1 = df_person[(df_person.address_street_name == "Franklin Ave") &(df_person.name.str.contains("Annabel"))].id.iloc[0]
suspect_id_2 = df_person[(df_person.address_street_name == "Northwestern Dr")].id.loc[df_person.address_number.idxmax()]
print(suspect_id_1,suspect_id_2)
df_person.info()
```

2.5 Merger les tableaux et trouver le tueur

```python
df_get_fit_now_member.head(5)
df_drivers_license.head(5)
df_get_fit_now_member =df_get_fit_now_member.astype({"id": "string", "name": "string","name":"string","membership_status": "cate
df_member_person = pd.merge(df_get_fit_now_member, df_person, left_on="person_id", right_on="id", how="inner",suffixes=("_member
df_member_person_with_license = pd.merge(df_member_person, df_drivers_license, left_on="license_id", right_on="id", how="inner",
# df_member_person_with_license.head(5)
# df_member_person_with_license.info()


tueur = df_member_person_with_license[(df_member_person_with_license['id_member'].str.startswith("48Z"))
                                      & (df_member_person_with_license['membership_status'] == "gold")]
print(tueur.iloc[0])
```

✓ 0.0s Python

```
id_member                              48Z55
person_id                              67318
name_member                    Jeremy Bowers
membership_start_date               20160101
membership_status                       gold
id_person                              67318
name_person                    Jeremy Bowers
license_id                            423327
address_number                           530
address_street_name      Washington Pl, Apt 3A
ssn                                871539279
id                                    423327
age                                       30
height                                    70
eye_color                              brown
hair_color                             brown
gender                                  male
plate_number                          0H42W2
car_make                           Chevrolet
car_model                           Spark LS
Name: 181, dtype: object
```